

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0641

Irving 138 kV
March 17, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0641. The Interconnection Customer proposes a 125.2 MW solar photovoltaic facility interconnecting at the Irving 138 kV substation in the Pennsylvania area. The request was received on July 07, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 243 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$91,745,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bullmoose to Morovis 8801 138.0 kV	W Gallatin to Walnut 345.0 kV	227.0	248	6.7
Morovis 8801 to Newcomerstown 138.0 kV	Walnut to W Gallatin 345.0 kV	200.6	289	6.7
Newcomerstown to Vly Home Tap 2 138.0 kV	West Gallatin to Walnut 345.0 kV Line	183.5	341	6.7
E Arlington to Valley Home Tap 2 138.0 kV	West Gallatin to Walnut 345.0 kV	178.0	362	6.7
Landis Delivery No 2 to East Arlington 138.0 kV Line	W Gallatin to Walnut 345.0 kV	135.2	225	7.4
Landis Delivery No 2 to Okawville 138.0 kV	West Gallatin Substation to Walnut 345.0 kV	135.1	232	7.4
Magic Kingdom Substation to Okawville 138.0 kV Line	W Gallatin to Walnut 345.0 kV	134.5	242	7.4
Coso Little Lae to Upsco 138.0 kV line	Upsco to Wheelabrator Frackville Energy 345.0 kV	133.6	180	8.5
Walnut to Clear Springs 345.0 kV	Tombstone to Bixby Hill 500.0 kV	132.4	1,101	5.7
Coso Little Lake to Locust Ridge II LLC 138.0 kV L	Wheelabrator Frackville Energy to Upsco 345.0 kV	125.5	204	8.5
Clear Springs 500 kV/345 kV Transformer	Curecanti to Moser Generating Station 345.0 kV	120.5	1,228	5.3
Locust Ridge II LLC to Drews 138.0 kV	Wheelabrator Frackville Energy Substation to Upsco 345.0 kV Line	116.2	234	8.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Magic Kingdom to Irving Substation 138.0 kV	Felida to W Gallatin 345.0 kV L	116.2	234	15.0
Wigwam to Upsco 345.0 kV line	Fayette to Ironwood Llc Substation 345.0 kV L	108.2	1,213	14.9
Upsco to Wheelabrator Frackville Energy 345.0 kV	W Gallatin to Walnut 345.0 kV	106.0	1,225	10.1
Rawhide to Bowen Street 138.0 kV	Morganville to Westwood Generation Llc 345.0 kV	104.1	350	5.9
W Gallatin to Flida 345.0 kV	Magic Kingdom to Irving 138.0 kV	104.0	1,554	5.1
Ontelaunee Energy Ctr Substation to Wigwam 345.0 kV Line	Fayette to Ironwood Llc 345.0 kV	103.7	1,096	14.8
Mercury Substation to Fayette 345.0 kV Line	Wigwam Substation to Upsco 345.0 kV Line	102.8	1,007	7.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Morovis 8801 to Bullmoose 138.0 kV	Rebuild line with higher-capacity conductor	\$1,206,000	100% (\$1,206,000)
Morovis 8801 to Newcomerstown 138.0 kV	Rebuild line with higher-capacity conductor	\$5,774,000	100% (\$5,774,000)
Newcomerstown to Vly Home Tap 2 138.0 kV	Rebuild line with higher-capacity conductor	\$6,155,000	100% (\$6,155,000)
E Arlington to Valley Home Tap 2 Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$3,148,000	100% (\$3,148,000)
E Arlington to Landis Delivery No 2 138.0 kV line	Rebuild line with higher-capacity conductor	\$6,465,000	100% (\$6,465,000)
Landis Delivery No 2 to Okawville 138.0 kV	Rebuild line with higher-capacity conductor	\$2,743,000	100% (\$2,743,000)
Okawville to Magic Kingdom 138.0 kV line	Rebuild line with higher-capacity conductor	\$8,898,000	100% (\$8,898,000)
Coso Little Lae to Upsco 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,296,000	100% (\$4,296,000)
Walnut to Clear Springs 345.0 kV	Rebuild line with higher-capacity conductor	\$50,163,000	2% (\$1,182,000)
Coso Little Lake to Locust Ridge li Llc 138.0 kV L	Rebuild line with higher-capacity conductor	\$16,434,000	100% (\$16,434,000)
Clear Springs 500/345kV Auto	Replace transformer with higher-rated unit	\$44,059,000	2% (\$743,000)
Locust Ridge li Llc to Drews 138.0 kV	Rebuild line with higher-capacity conductor	\$5,856,000	100% (\$5,856,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Magic Kingdom to Irving Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$26,249,000	100% (\$26,249,000)
Wigwam to Upsco Substation 345.0 kV	Reconductor line	\$24,886,000	5% (\$1,269,000)
Upsco to Wheelabrator Frackville Energy 345.0 kV	Reconductor line	\$11,020,000	4% (\$426,000)
Rawhide to Bowen Street 138.0 kV	Reconductor line	\$1,091,000	16% (\$180,000)
Felida to W Gallatin 345.0 kV	Reconductor line	\$9,937,000	2% (\$218,000)
Ontelaunee Energy Ctr Substation to Wigwam 345.0 kV Line	Reconductor line	\$6,174,000	5% (\$290,000)
Mercury Substation to Fayette 345.0 kV Line	Reconductor line	\$7,235,000	3% (\$213,000)
	Total allocated to project		\$91,745,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.